

R3: Noise-Floor Mis-specification

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How accurately must the sensor noise PSD be known?

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SHARC buoy project Code: `runR3NoiseMisspecification.m` Results: `Results/R3/R3_20261002_124033`

Summary

Question. The transmissibility estimator subtracts an assumed reference-noise PSD. If that assumption is wrong by a fraction ϵ , how biased are β and R ? **Outcome (LSM6DSV16X).** The noise floor must be known to within $\pm 30.6\%$ to keep the bias below half a Fisher uncertainty unit ($d_{\text{sys}} \leq 0.5$), or $\pm 61\%$ for $d_{\text{sys}} \leq 1$. Omitting the correction altogether gives $d_{\text{sys}} = 1.63$, so **the correction is needed**, even though the stiffness error it causes is only -0.11% . The measured colour of the noise is irrelevant: a flat band-mean PSD in place of the measured spectrum gives $d_{\text{sys}} = 0.025$. **Controlling quantity.** The reference-channel SNR: the tolerance tightens from “not needed” (datasheet grade, $\text{SNR} \geq 225$) to $\pm 7.7\%$ (LSM6DS3TR-C, $\text{SNR} \approx 1$).

1 What the study does

Why only the reference noise matters. The estimator is

$$\hat{T}_j = \frac{\hat{S}_{jr}}{\hat{S}_{rr} - S_{nn,\text{assumed}}}.$$

Noise on a non-reference sensor is uncorrelated with the reference, so it averages out of $\hat{S}_{jr}$ and adds variance but no bias. Noise on the reference enters $\hat{S}_{rr}$ in expectation and must be subtracted. An error in that subtraction biases every transmissibility by the same factor.

The bias. With $S_{nn,\text{assumed}} = (1 + \epsilon)S_{nn}$, the error is $\Delta = \epsilon S_{nn}$ ($\epsilon = -100\%$ is no correction). Using the window-smoothed (Welch-expected) spectra, $T_W = S_{jr,W}/S_{rr,\text{sig},W}$, the expected estimate becomes $T_W S_{rr,\text{sig},W}/(S_{rr,\text{sig},W} - \Delta)$, so the additional feature bias is

$$b_{\text{noise}} = T_W \frac{\Delta}{S_{rr,\text{sig},W} - \Delta} \approx T_W \epsilon \frac{S_{nn}}{S_{rr,\text{sig}}} \quad (\text{small } \epsilon).$$

This is exact in expectation for a fixed bin set; the first-order form shows that the sensitivity is set by the noise-to-signal ratio of the reference. Overestimating the floor ($\epsilon > 0$) shrinks the denominator and diverges as $\Delta \rightarrow S_{rr,\text{sig}}$, so positive errors grow faster.

Projection. As in R4, the E1 engine maps the bias to $\delta\theta = F^{-1}J^T\Sigma^{-1}b_{\text{noise}}$, $d_{\text{sys}} = \sqrt{\delta\theta^T F \delta\theta}$ and $\delta\beta/\beta = e^{\delta \ln \beta} - 1$. For the class, the bias adds as a vector to the Welch bias: $\delta\theta_{\text{tot}} = \delta\theta_W + \delta\theta_{\text{noise}}$. Setting: p_0 , Level 2C, $N = 65536$, $L = 2048$, JONSWAP $H_s = 0.05$ m, all six noise cases; pre-registered $\epsilon \in \{-100, -50, -30, -10, +10, +30, +50\}\%$. Tolerances are first crossings on a fine grid over $[-100\%, +50\%]$ (1501 points), taking the smaller $|\epsilon|$ of the two signs. No new EMM solves were needed.

Checks. The bias passed through `evaluateScenario`’s own systematic-bias path gives the same d_{sys} (0.490723 at $\epsilon = +30\%$, LSM6DSV16X; relative difference 0). The coloured case (pre-registered for the LSM6DS3TR-C, also run for the other two measured sensors) uses the *measured* basin-band noise shape, a power law fitted to the 16.9 h static record, while the estimator subtracts its flat band mean.

2 Results

Required noise-floor accuracy (smaller $|\epsilon|$ of the two signs; search range $[-100\%, +50\%]$).

Noise case	S_{nn}	min SNR $S_{rr,\text{sig}}/S_{nn}$	tolerance $ \epsilon $			no correction ($\epsilon = -100\%$)	
			$ \delta\beta/\beta \leq 1\%$	$d_{\text{sys}} \leq 0.5$	$d_{\text{sys}} \leq 1$	d_{sys}	$\delta\beta/\beta$
Datasheet grade	6.5×10^{-8}	225	n.r.	n.r.	n.r.	0.39	-0.005%
Three IMUs fused	5.0×10^{-7}	29	n.r.	39.3%	78.7%	1.27	-0.063%
LSM6DSV16X	8.0×10^{-7}	18	n.r.	30.6%	61.3%	1.63	-0.106%
G3 assumption	1.42×10^{-6}	10	n.r.	22.7%	45.4%	2.19	-0.194%
ICM-42688-P	1.52×10^{-6}	10	n.r.	21.9%	43.7%	2.27	-0.209%
LSM6DS3TR-C	1.24×10^{-5}	1	57.6%	7.7%	15.5%	6.25	-1.70%

n.r.: not reached within $[-100\%, +50\%]$ (not “larger than 100%”). min SNR: smallest reference signal-to-noise ratio over the accepted bins.

Coloured noise (true = measured shape; assumed = flat band mean).

Sensor	Fitted PSD shape	Max $ \Delta /S_{nn}$	$\delta\beta/\beta$	d_{sys}	d_{tot} (vs d_W alone)
LSM6DSV16X	$\omega^{-0.15}$	0.09	$+0.0031\%$	0.025	0.308 (0.309)
ICM-42688-P	$\omega^{-0.06}$	0.03	$+0.0023\%$	0.013	0.235 (0.236)
LSM6DS3TR-C (pre-registered)	$\omega^{-0.10}$	0.05	$+0.0315\%$	0.059	0.100 (0.088)

Smoothed measured spectra lie within 7.5 to 9.4% of their power-law fits; fitted band means match the configured S_{nn} to within 1%.

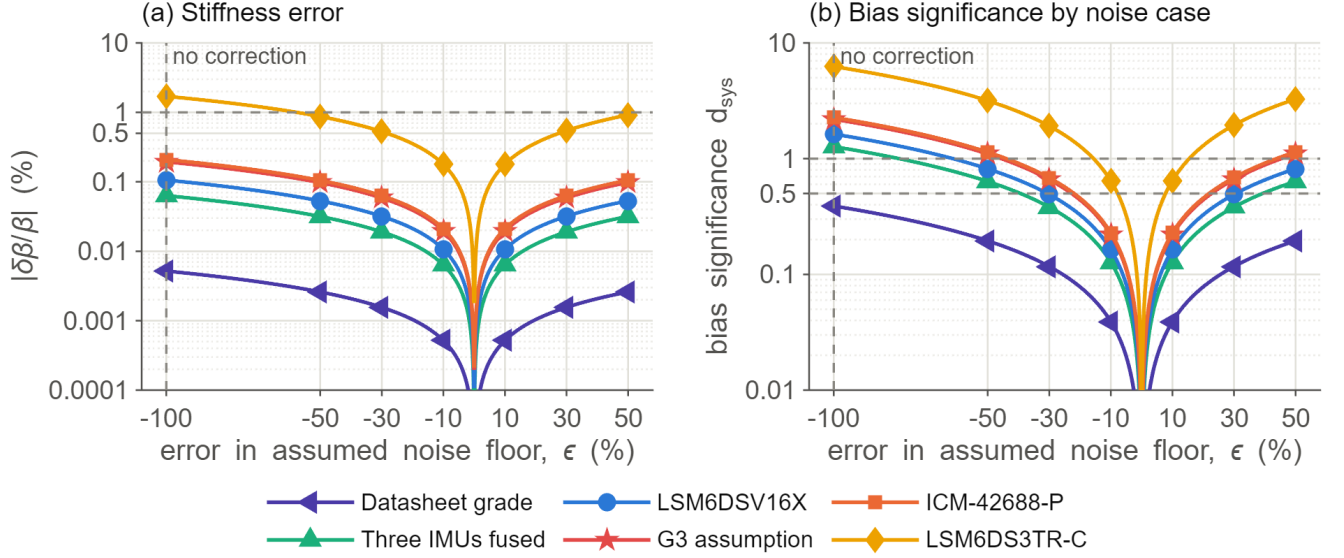


Figure 1: Effect of reference noise-floor mis-specification on the recovered parameters. (a) Magnitude of the induced stiffness error, $|\delta\beta/\beta|$, and (b) systematic bias significance, d_{sys} , as functions of the fractional error ϵ in the assumed reference noise PSD for all six noise cases. Solid curves show the fine tolerance sweep and markers indicate the pre-registered values of ϵ . The horizontal dashed lines indicate the 1% stiffness-error threshold in (a) and $d_{\text{sys}} = 0.5$ and 1 in (b). The vertical dashed line at $\epsilon = -100\%$ corresponds to applying no reference-noise correction.

3 Interpretation

1. **Correction is required, precise calibration is not.** For the LSM6DSV16X the noise floor must be subtracted (no correction: $d_{\text{sys}} = 1.63$), but a band-mean value good to about $\pm 30\%$ is enough. The statistical uncertainty of the band mean from the 16.9 h static record is far below this.
2. **Small physical error, significant statistical error.** Without correction the stiffness moves by only -0.11% , yet $d_{\text{sys}} = 1.63$: the estimator is precise enough ($\sigma_{\ln \beta} = 0.36\%$) that a 0.1% -scale shift, together with the corresponding shift in R , is large relative to its uncertainty ellipse. The 1% criterion is reached only for the LSM6DS3TR-C. As in R4, the significance criterion is the binding one.
3. **A simple rule.** The bias is nearly linear in ϵ for $|\epsilon| \lesssim 50\%$, so $d_{\text{sys}}(\epsilon) \approx |\epsilon| d_{\text{sys}}(\text{no correction})$ and

$$|\epsilon|_{d=0.5} \approx \frac{0.5}{d_{\text{sys}}(\text{no correction})}$$

(LSM6DSV16X: $0.5/1.63 = 30.7\%$ against 30.6% ; LSM6DS3TR-C: 8.0% against 7.7%). One evaluation without correction therefore gives the tolerance for any sensor or floe.

4. **Asymmetry.** Positive errors (over-subtraction) grow faster than negative ones, because the corrected denominator approaches zero in the lowest-SNR bins; this is visible on the right of Figure 1. The tolerance takes the smaller of the two signs.
5. **Spectral colour does not matter.** The measured noise spectra are nearly flat over the band ($\omega^{-0.06}$ to $\omega^{-0.15}$). Replacing them by their flat band mean adds at most $d_{\text{sys}} = 0.06$, and d_{tot} is essentially the Welch value. A frequency-dependent noise model is not needed for the production estimator.

Limitations.

- **SNR depends on the forcing.** The tolerance scales with $S_{nn}/S_{rr,\text{sig}}$. In calmer conditions than $H_s = 0.05$ m the SNR falls and the tolerance tightens: halving the wave height cuts $S_{rr,\text{sig}}$ by four, which quadruples the bias; the tolerance tightens by somewhat less than four because σ also grows. R1 and R2 should be read together with this.
- **Lab versus deployment noise.** S_{nn} was measured in a static lab test. Mounting, temperature and platform vibration can change it in the field; the $\pm 30\%$ margin is the budget for that.
- **Fixed bin set and covariance.** Bins accepted under the true noise are kept; the fraction that would flip under the wrong correction is in R3_sweep.csv. Σ_k uses the true white level; the coloured case is a bias test only.
- **One floe.** p_0 , Level 2C, one sea state.

Requirement for the experiment and the thesis

Keep the reference-noise correction, using the flat band-mean S_{nn} of the reference sensor. For the LSM6DSV16X at p_0 its value must be within about $\pm 30\%$ of the true in-situ level: the static calibration easily achieves this, so the practical requirement is to confirm that the deployed noise level has not changed by more than that. Noisier sensors need proportionally tighter knowledge ($\pm 7.7\%$ for the LSM6DS3TR-C).

Outputs (Results/R3/): R3_20261002.124033.log/.mat/_summary.txt; R3_sweep.csv (noise case $\times \epsilon$: $\delta\beta$, δR , d_{sys} , d_{tot} , flipped-bin fraction, class); R3_tolerance.csv; R3_coloured.csv; figures/R3_fig1_noise_mispecification (.pdf/.png).